

HandsOn-19

As an intern at OpenAI you are assigned the task of creating and testing a React application which will fetch and display a list of repository names for a given user.

Implementation

→App.js

```
import React, { useState } from 'react';
import GitClient from './GitClient';

function App() {
  const [username, setUsername] = useState("");
  const [repositories, setRepositories] = useState([]);
  const [error, setError] = useState("");
  const handleFetchRepos = async () => {
    if (!username) {
      setError('Username is required!');
      setRepositories([]);
      return;
    }
    setError("");
    const client = new GitClient();
    try {
      const repos = await
client.getRepositories(username);
      setRepositories(repos);
    } catch (err) {
      setError('Failed to fetch repositories.');
```

```

<div style={{ padding: '30px', fontFamily: 'Arial' }}>
  <h2>GitHub Repository Viewer</h2>
  <input
    type="text"
    value={username}
    onChange={(e) => setUsername(e.target.value)}
    placeholder="Enter GitHub username"
    style={{ padding: '8px', marginRight: '10px',
width: '250px' }}
  />
  <button onClick={handleFetchRepos} style={{
padding: '8px 16px' }}>
    Fetch Repos
  </button>
  {error && <p style={{ color: 'red', marginTop:
'10px' }}>{error}</p>}
  <ul style={{ marginTop: '20px' }}>
    {repositories.map((repo, index) => (
      <li key={index}>{repo}</li>
    ))}
  </ul>
</div>
);
}
export default App;

```

→ App.css

```

.App {
  text-align: center;
}
.App-logo {

```

```
height: 40vmin;
pointer-events: none;
}
@media (prefers-reduced-motion: no-preference) {
  .App-logo {
    animation: App-logo-spin infinite 20s linear;
  }
}
.App-header {
  background-color: #282c34;
  min-height: 100vh;
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: center;
  font-size: calc(10px + 2vmin);
  color: white;
}
.App-link {
  color: #61dafb;
}
@keyframes App-logo-spin {
  from {
    transform: rotate(0deg);
  }
  to {
    transform: rotate(360deg);
  }
}
```

→GitClient.js

```
import axios from 'axios';

class GitClient {

  async getRepositories(username) {

    const response = await axios.get(`https://api.github.com/users/${username}/repos`);

    return response.data.map(repo => repo.name);

  }

}

export default GitClient;
```

→CohortDetails.js

```
// src/CohortDetails.js
import React from 'react';

const CohortDetails = ({ cohort }) => {
  return (
    <div className="cohort">
      <h3>{cohort.code}</h3>
      <p>{cohort.name}</p>
      <p>{cohort.status}</p>
    </div>
  );
};

export default CohortDetails;
```

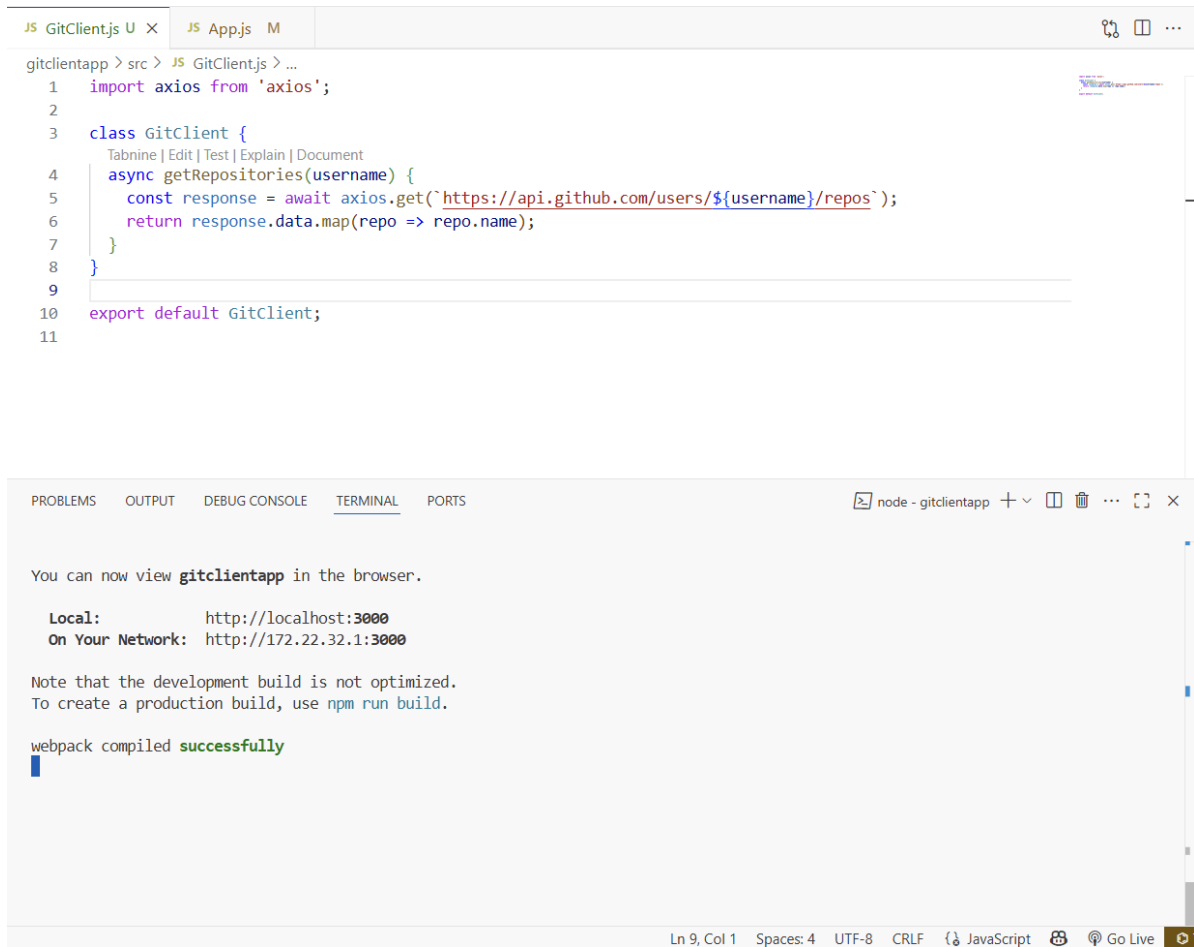
→GitClient.test.js

```
import axios from 'axios';
import GitClient from './GitClient';

jest.mock('axios');

describe('Git Client Tests', () => {
  test('should return repository names for techiesyed', async () => {
    const mockedData = [{ name: 'repo1' }, { name: 'repo2' }];
    axios.get.mockResolvedValue({ data: mockedData });
    const client = new GitClient();
    const repos = await client.getRepositories('techiesyed');
    expect(repos).toEqual(['repo1', 'repo2']);
    expect(axios.get).toHaveBeenCalledWith('https://api.github.com/users/techiesyed/repos');
  });
});
```

Output



The screenshot shows a VS Code editor with a file named `GitClient.js` open. The code defines a class `GitClient` with an asynchronous method `getRepositories(username)` that uses `axios` to fetch repositories from GitHub. The terminal output shows the application running successfully on `localhost:3000`.

```
gitclientapp > src > JS GitClientjs > ...
1  import axios from 'axios';
2
3  class GitClient {
4      Tabnine | Edit | Test | Explain | Document
5      async getRepositories(username) {
6          const response = await axios.get(`https://api.github.com/users/${username}/repos`);
7          return response.data.map(repo => repo.name);
8      }
9
10 export default GitClient;
11
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

You can now view **gitclientapp** in the browser.

Local: http://localhost:3000
On Your Network: http://172.22.32.1:3000

Note that the development build is not optimized.
To create a production build, use `npm run build`.

webpack compiled **successfully**

Ln 9, Col 1 Spaces: 4 UTF-8 CRLF JavaScript Go Live



Repositories of techiesyed

- ArrayListDemo
- ArrayListDemo01
- AzureDevopsDemoProductsApi
- CICDMvcDemo
- CleanArchitecture
- ContosoUniversity
- DelegatesDemo
- EventsDemo
- FISGlobal-0721
- GenericCollections
- GenericsDemo
- GitHubDemoApp
- GitTestFolder
- HashTableDemo
- HelloJenkins
- HelloWorldJenkinsAngular
- Layered-Architecture-Demo
- musical-octo-carnival
- mvc03
- Mvvm-With-WPF
- MyApi
- ReflectionDemo
- stunning-broccoli
- techiesyed
- TestRepo
- Webinar-Automation-testing-Selenium-CSharp
- WhyUseGenerics

